Mathematics practice booklet: Grade 6 expectations

Grade 6 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 6 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 6.EE.A.1, 6.EE.A.2, 6.EE.A.3, 6.EE.A.4, 6.EE.B.5, 6.EE.B.6, 6.EE.B.7, 6.EE.B.8, 6.EE.C.9

A worked example

A falcon eats 5 portions of food a day, so 5x = 35 when x is the number of days a box lasts. Solve for x, then work out 2x + 3.

1. 5x means 5 times x. To undo the multiplying, divide both sides by 5.
2. x = 35 / 5 = 7.
3. Now replace x with 7 in 2x + 3: 2 x 7 + 3 = 14 + 3 = 17.

x = 7; 2x + 3 = 17

Questions

1. Evaluate: 3^2 + 4 x 2

   A. 14

   B. 13

   C. 17

   D. 26

2. Evaluate: 2^3 x 5

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3. Which expression means five less than twice a number n?

   A. 5 - 2n

   B. 2n - 5

   C. 2(n - 5)

   D. n - 10

4. A taxi charges AED 4 to start and AED 3 for each kilometre, so a trip of x km costs 3x + 4 dirhams. Find the cost of a 6 km trip.

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5. Are 2(x + 3) and 2x + 6 equivalent expressions? Explain.

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6. Simplify: 4a + 3a

   A. 7a

   B. 12a

   C. 7a^2

   D. 43a

7. Solve: x + 9 = 25

   A. 34

   B. 9

   C. 225

   D. 16

8. Noura shares 48 date cookies equally among 6 plates. Write the equation 6n = 48 and solve it to find n, the cookies on each plate.

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9. Solve: n / 4 = 7

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10. Dates cost AED 12 a kilogram at the souq. Maryam pays AED 60. Write an equation for the number of kilograms k and solve it.

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11. A ride at a fun park needs a child to be taller than 120 cm. Which inequality shows the height h in cm?

    A. h < 120

    B. h ≥ 120

    C. h > 120

    D. h = 120

12. A car park charges AED 4 for each hour. Let h be the hours and c the cost in AED. Write an equation, say which variable is independent, and find the cost for 5 hours.

Section 2: Shapes, measuring and space

Standards practised: 6.G.A.1, 6.G.A.2, 6.G.A.3, 6.G.A.4

A worked example

A triangular shade sail over a play area has a base of 8 m and a height of 5 m. What is its area?

1. A triangle is half of a rectangle with the same base and height.
2. The rectangle would be 8 x 5 = 40 square metres.
3. Half of 40 is 20.

20 square metres

Questions

1. Find the area of a triangle with base 10 cm and height 6 cm.

   A. 60 square cm

   B. 16 square cm

   C. 30 square cm

   D. 32 square cm

2. A flower bed in a park is a parallelogram with base 8 m and height 5 m. What is its area?

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3. A garden is shaped like an L. Split it into a rectangle 8 m by 5 m and a rectangle 4 m by 3 m. What is the total area?

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4. A triangle has an area of 24 square cm and a base of 8 cm. What is its height?

   A. 3 cm

   B. 12 cm

   C. 24 cm

   D. 6 cm

5. Hamdan fills a rectangular water tank that is 5/2 m long, 2 m wide and 3/2 m high. How many cubic metres of water does it hold when full?

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6. A cube has edges of 1/2 cm. What is its volume?

   A. 1/8 cubic cm

   B. 1/6 cubic cm

   C. 3/2 cubic cm

   D. 1/2 cubic cm

7. A rectangle has corners at (1, 2), (6, 2), (6, 5) and (1, 5). Find its length, its width and its area.

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8. Two points are at (-3, 4) and (-3, -2). How far apart are they?

   A. 2 units

   B. 6 units

   C. 7 units

   D. 12 units

9. A net is folded to make a rectangular box. What faces does the net have?

   A. 4 rectangles

   B. 6 triangles

   C. 6 rectangles

   D. 8 rectangles

10. Hind wraps a gift box that is 4 cm long, 3 cm wide and 2 cm high. The paper does not overlap. How much paper covers the box?

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11. A square pyramid has a square base with sides 6 cm and 4 triangular faces, each with base 6 cm and height 5 cm. Find its surface area.

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12. A triangle and a parallelogram have the same base and the same height. How do their areas compare? Explain.

Section 3: Data, graphs and chance

Standards practised: 6.SP.A.1, 6.SP.A.2, 6.SP.A.3, 6.SP.B.4, 6.SP.B.5

A worked example

Five pupils read 3, 5, 8, 4 and 10 books this term. Find the mean and the median.

1. Mean: add the values, 3 + 5 + 8 + 4 + 10 = 30, then divide by 5 to get 6.
2. Median: put them in order, 3, 4, 5, 8, 10.
3. The middle value is 5.

Mean 6; median 5

Questions

1. Which of these is a statistical question?

   A. How many hours did Salem sleep last night?

   B. How many hours do the pupils in Grade 6 usually sleep?

   C. How many pupils are in Salem's class today?

   D. What day is it today?

2. Khalid walks 4, 6, 7, 9 and 9 km on five days. Find the mean distance he walks in a day.

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3. Find the median of 12, 5, 9, 7, 15, 3.

   A. 8

   B. 8.5

   C. 9

   D. 7

4. Find the range of 3, 5, 7, 9, 12, 15.

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5. Find the mean absolute deviation (MAD) of 3, 5, 7, 9.

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6. Find the interquartile range (IQR) of 4, 6, 7, 9, 10, 12, 15.

   A. 11

   B. 9

   C. 3

   D. 6

7. A box plot of taxi fares in AED has minimum 20, lower quartile 30, median 35, upper quartile 50 and maximum 80. What are the range and the IQR?

   A. Range 60; IQR 15

   B. Range 80; IQR 50

   C. Range 60; IQR 20

   D. Range 50; IQR 20

8. A dot plot shows books read by 15 pupils: 1 book (2 dots), 2 books (4 dots), 3 books (5 dots), 4 books (3 dots), 5 books (1 dot). What is the mode?

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9. Using the same dot plot, what is the median number of books?

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10. A histogram shows minutes spent on homework: 0 to 9 minutes (4 pupils), 10 to 19 minutes (7 pupils), 20 to 29 minutes (5 pupils). How many pupils are there, and how many spent less than 20 minutes?

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11. Waiting times in minutes at a metro station on five days: 2, 3, 3, 4, 30. Which measure better describes a typical wait, and why?

    A. The mean, because it uses every value

    B. The median, because one very long wait pulls the mean up

    C. The mean, because it is the largest number

    D. The median, because it is the largest value

12. The values 6, 8, 10 have mean 8 and median 8. A value of 20 is added. What happens to the mean and the median?

Section 4: Numbers and calculating

Standards practised: 6.RP.A.1, 6.RP.A.2, 6.RP.A.3, 6.NS.A.1, 6.NS.B.2, 6.NS.B.3, 6.NS.B.4, 6.NS.C.5, 6.NS.C.6, 6.NS.C.7

A worked example

Work out 3/4 divided by 1/2. Then find 20% of 60.

1. To divide by a fraction, multiply by its reciprocal: 3/4 x 2/1 = 6/4 = 3/2 = 1 1/2.
2. 20% means 20 out of 100, or 1/5.
3. One fifth of 60 is 12.

1 1/2 and 12

Questions

1. A tray has 12 red and 8 white tiles. What is the ratio of red to white in simplest form?

   A. 2:3

   B. 12:8

   C. 6:4

   D. 3:2

2. 5 kg of dates cost AED 90. What is the unit rate, the cost for 1 kg?

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3. A recipe uses 2 cups of rice for every 3 cups of water. How much water is needed for 8 cups of rice?

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4. A shop takes 25% off a bag that costs AED 80. How many dirhams is the discount?

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5. 30% of a number is 45. What is the number?

   A. 13.5

   B. 150

   C. 75

   D. 1,350

6. A metro train travels 90 km in 1.5 hours at a steady speed. How far does it go in 1 hour?

   A. 135 km

   B. 45 km

   C. 60 km

   D. 90 km

7. Layla has 3/4 litre of juice and pours it into small cups that each hold 1/8 litre. How many cups does she fill?

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8. Divide: 2 1/2 / 5/6

   A. 3

   B. 12/25

   C. 2 1/12

   D. 5/3

9. Maryam buys 3.5 kg of dates at AED 12.6 for each kilogram. How much does she pay?

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10. Calculate: 7.56 / 0.6

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11. What are the greatest common factor (GCF) and the least common multiple (LCM) of 12 and 18?

    A. GCF 36; LCM 6

    B. GCF 3; LCM 36

    C. GCF 6; LCM 216

    D. GCF 6; LCM 36

12. Put in order from least to greatest: -2.5, 1/2, -4, 0, 1.5. Then say which is colder, -8 degrees or -3 degrees, and which of the two has the greater absolute value.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. C. 17. 3^2 = 9 and 4 x 2 = 8, so 9 + 8 = 17.
2. 40 (2^3 = 2 x 2 x 2 = 8, and 8 x 5 = 40)
3. B. 2n - 5. Twice n is 2n, and five less than that is 2n - 5.
4. AED 22 (3 x 6 = 18, and 18 + 4 = 22)
5. Yes: 2 multiplies both x and 3, so 2(x + 3) = 2x + 6 for every value of x.
6. A. 7a. 4 of something plus 3 of the same thing is 7a.
7. D. 16. Undo adding 9 by subtracting 9 from both sides: x = 25 - 9 = 16.
8. n = 8 cookies on each plate (48 / 6)
9. n = 28 (7 x 4)
10. 12k = 60, so k = 5 kilograms.
11. C. h > 120. Taller than 120 means h is greater than 120, so h > 120 (120 itself does not work).
12. c = 4h; h is independent (you choose it) and c depends on it; 5 hours cost AED 20.

Section 2: Shapes, measuring and space

1. C. 30 square cm. A triangle is half of a rectangle: 10 x 6 = 60, and half of 60 is 30 square cm.
2. 40 square metres (8 x 5)
3. 52 square metres (40 + 12)
4. D. 6 cm. 24 is half of 8 x h, so 8 x h = 48 and h = 6 cm.
5. 15/2 = 7 1/2 cubic metres (5/2 x 2 x 3/2)
6. A. 1/8 cubic cm. Volume is 1/2 x 1/2 x 1/2 = 1/8 cubic cm.
7. Length 5 units, width 3 units, area 15 square units.
8. B. 6 units. They share the same x value. 4 is above zero and 2 is below zero, so 4 + 2 = 6 units.
9. C. 6 rectangles. A rectangular box has 6 faces, and each one is a rectangle.
10. 52 square cm (2 x 12 + 2 x 8 + 2 x 6 = 24 + 16 + 12)
11. 96 square cm (base 36 + four triangles of 15 each = 36 + 60)
12. The triangle has half the area: two copies of the triangle fit together to make the parallelogram.

Section 3: Data, graphs and chance

1. B. How many hours do the pupils in Grade 6 usually sleep? It expects answers that vary from pupil to pupil. The others have one fixed answer.
2. 7 km (35 / 5)
3. A. 8. In order: 3, 5, 7, 9, 12, 15. The middle two are 7 and 9, and (7 + 9) / 2 = 8. (8.5 is the mean.)
4. 12 (15 - 3)
5. 2 (mean 6; distances 3, 1, 1, 3 add to 8; 8 / 4 = 2)
6. D. 6. Median 9; lower half 4, 6, 7 has median 6; upper half 10, 12, 15 has median 12; 12 - 6 = 6. (11 is the range.)

7. C. Range 60; IQR 20. Range is 80 - 20 = 60. IQR is 50 - 30 = 20.
8. 3 books (it has the most dots)
9. 3 books (the 8th of 15 values; 2 + 4 = 6 values are 2 or fewer, and the next 5 are 3)
10. 16 pupils in all; 11 spent less than 20 minutes (4 + 7).
11. B. The median, because one very long wait pulls the mean up. The median is 3 minutes. The one wait of 30 pulls the mean up to 8.4, which is not typical.
12. Mean becomes 11 and median becomes 9. Both rise, but the mean rises more because it is pulled by the large value.

Section 4: Numbers and calculating

1. D. 3:2. Divide both numbers by 4: 12:8 = 3:2.
2. AED 18 per kg (90 / 5)
3. 12 cups (8 is 4 times 2, so 3 x 4 = 12)
4. AED 20 (25% is 1/4, and 80 / 4 = 20)
5. B. 150. 10% is 45 / 3 = 15, so 100% is 150.
6. C. 60 km. 90 / 1.5 = 60 km.
7. 6 cups (3/4 x 8/1 = 24/4)
8. A. 3. 2 1/2 = 5/2, and 5/2 x 6/5 = 30/10 = 3.
9. AED 44.1 (12.6 x 3.5)
10. 12.6 (the same as 75.6 / 6)
11. D. GCF 6; LCM 36. Factors in common: 1, 2, 3, 6, so the GCF is 6. Multiples of 18: 18, 36 (12 x 3), so the LCM is 36.
12. -4, -2.5, 0, 1/2, 1.5. -8 degrees is colder, and -8 also has the greater absolute value (8 is greater than 3).